



Fantasies embedded

Embedded systems and real time operating systems are fast achieving ubiquity, blurring the lines between science fiction and hard reality

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A doctor is configuring a cardiac pacemaker inside his patient's chest while sitting 200 kilometres away. Another person is travelling in a driver-less car that takes him from Mumbai to New Delhi using its in-built navigation program. Impossible? Not really. Advances in technology have taken place at such speed that these fictitious scenarios are likely to be translated into reality within the next couple of years.

Real Time Operating Systems (RTOS) and embedded systems are two among the several technologies that will play a major role in making these concepts possible.

You may say these technologies have been around for a long time and there have been no such dramatic developments. But what you may not know is that RTOS and embedded systems are the unsung heroes of much of the technology you use today—the video game you play, or the CD player or the washing machine you use employ them. Without an embedded system and an RTOS, you would not even be able to go online

using your modem. Read on to find out what these technologies have in store for you.

How 'real' is real time?

When it comes to real-life situations like controlling a traffic signal or a steel furnace, the OS has to respond immediately. Imagine the plight of a driver if his car's anti-skid mechanism refuses to respond the moment he jams his brakes. This is where RTOS comes in. Real time operating systems are those that respond to inputs immediately (*see box, 'What is an RTOS?'*).

First step—embedded technology

Why use a hammer when a gentle touch would do? This is the basic idea behind all the appliances that make use of embedded systems. They are pre-programmed to perform a dedicated or narrow range of functions as part of a larger system, usually with minimal end-user interaction. For example, an entire operating system like Windows 98 is not required for controlling a car's valve that has only certain types of movements, or